The CBH theorem

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Goal: Distinguish between classical and quantum theory in terms of their distinctive information-theoretic characteristics.

Constraint: To **prove** a characterization theorem, we must assume a background mathematical framework that is general enough to accommodate both classical and quantum sytems.

classical mechanics with C^*-algebras

If X is the phase space of a classical system, then its observables are represented by elements of $C(X)$, the continuous functions from X to $\mathbb{C}$ that are constant at infinity.

$C(X)$ has pointwise sums, products, and complex conjugates. Equipped with the supremum norm, it is an **abelian** or **commutative** C^*-algebra.

The pure states of $C(X)$ correspond to points in X. The mixed states of $C(X)$ correspond to probability measures on X.

composite classical systems

If Y is the phase space of another classical system, then:

$$C(X \times Y) \simeq C(X) \otimes C(Y).$$

Fact: No states are entangled across $C(X) \otimes C(Y)$. The Bell inequality is satisfied in all states. (Easy proof using the fact that $C(X)$ and $C(Y)$ are abelian.)

cloning

The system $\mathcal{A} \otimes \mathcal{A}$ allows **cloning** just in case there is a state ω of $\mathcal{A}$ and a CP map $T : \mathcal{A} \otimes \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ such that

$$T^*(\rho \otimes \omega) \;=\; \rho \otimes \rho,$$

for all states ρ of $\mathcal{A}$.

Fact: The system $C(X) \otimes C(X)$ allows cloning.

Proof: Verify that any homomorphism $f : Y \to Z$ gives rise to a CP map $f^* : C(Z) \to C(Y)$. Then note that the map $f(\langle x, z \rangle) = \langle x, x \rangle$ is a homomorphism.

all classical systems allow cloning

According to the **Gelfand representation theorem**, a C^*-algebra $\mathcal{A}$ is abelian iff $\mathcal{A} \simeq C(X)$ for some topological space X.

Corollary: If $\mathcal{A}$ is commutative then $\mathcal{A} \otimes \mathcal{A}$ allows cloning.

no quantum system allows cloning

New use for an old concept: If K is a convex set, and $x, y \in K$, then the **affine ratio** between x, y, written $p(x|y)$, is the largest value of $f(y)$ where f ...

Fact: The set of states of a classical system is a simplex; hence the affine ratio between any two pure states is 0.

Fact: If x, y are quantum states with affine ratio 0, then there is a continuous path of quantum states from x to y, achieving all intermediate affine ratios.